TSN END POINT PCIe NIC

The QBIT TSN End Point PCIe NIC(QSXG2-P) is a Time-Sensitive Networking (TSN) endpoint interface designed for deterministic Ethernet deployments. It integrates dual 100/1000BASE-SFP TSN Ethernet ports and a PCIe host interface for attachment to a computer, providing a turnkey path to add TSN connectivity to host-based systems.

The NIC combines Layer-2 features such as VLAN tagging, QoS, multicast control and spanning tree with key IEEE 802.1 TSN standards for timing synchronization, traffic shaping, scheduled traffic, per-stream policing and reliability. Secure configuration and lifecycle management are available through HTTPS, SSHv2 and Netconf/YANG, with encrypted and digitally signed firmware/bitstream upgrades.

Introduction

Time-Sensitive Networking (TSN) extends standard Ethernet with deterministic latency, bounded jitter and precise time synchronization. The TSN End Point PCIe NIC enables industrial PCs and embedded hosts to participate as TSN endpoints in converged Ethernet networks.

It supports IEEE 802.1AS gPTP timing, credit-based and time-aware traffic shaping, per-stream filtering and policing, and frame replication/elimination for reliability. Comprehensive Layer-2 features and secure management interfaces simplify deployment in mission-critical networks.

Architecture

The Qbit TSN Endpoint Card is a PCIe-based network interface that brings full hardware-accelerated Time-Sensitive Networking (TSN) to a host system. It supports deterministic, low-latency Ethernet communication through dedicated transmit and receive datapaths.

On the transmit side, packets from the host pass through PCIe DMA into traffic buffers, undergo Credit-Based Shaping (802.1Qav), optional Frame Replication for redundancy (802.1CB), time-aware scheduling, frame preemption (802.1Qbu), and precise egress timestamping before reaching the MAC and physical ports.

On the receive path, incoming frames are timestamped at ingress, filtered and identified by stream, processed for duplicate elimination via FRER recovery, and then buffered for efficient DMA transfer to the host.

A central gPTP state machine (802.1AS) maintains sub-microsecond network synchronization, supported by a high-precision timestamp generator and 1 PPS output via an SMA connector.

Register-based configuration from the host allows flexible control of streams, queues, redundancy rules, and scheduling parameters. This offloaded architecture minimizes software involvement, ensuring bounded latency, fault tolerance, and precise timing for real-time industrial, automotive, and AVB applications.

Why Choose Us?

🚀 Rapid integration, faster time-to-market

📘 Clean, verified, documented RTL

🤝 Direct access to experts

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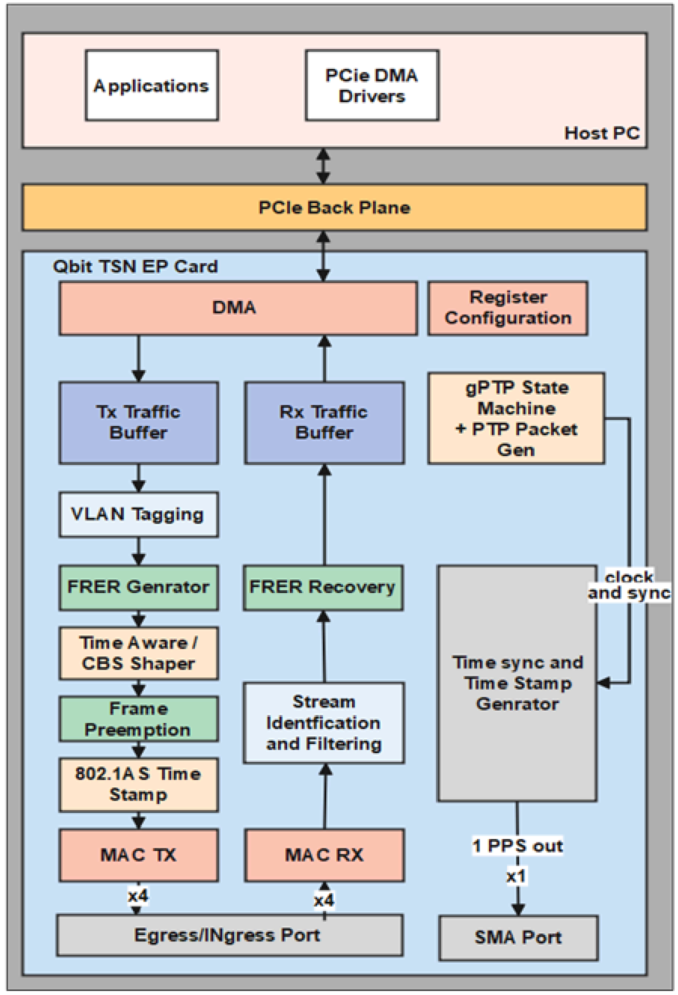
Features

- **2x 10/100/1000BASE-T** TSN Ethernet ports for deterministic endpoint connectivity.
- **1x PCIe host interface** for attachment to a host computer.
- **IEEE 802.3-2008** Ethernet compliant.
- **Automatic MAC address** learning and aging with support for static MAC table entries.
- **Port-based VLANs with IEEE 802.1Q** VLAN tagging support (up to 4K VLAN groups).
- **IEEE 802.1p** Class of Service (CoS) / Quality of Service (QoS) support.
- **IEEE 802.1AB Link Layer** Discovery Protocol (LLDP) support.
- **Port rate limiting and storm control** for flooded broadcast, multicast and unicast traffic.
- **Layer 2 multicast filtering** with IGMP snooping (up to 1024 multicast filters).
- **Spanning tree support:** STP (IEEE 802.1D), RSTP (IEEE 802.1w) and MSTP (IEEE 802.1s).
- **IEEE 802.1AS** timing and synchronization (gPTP).
- **IEEE 802.1Qav** Credit Based Shaper (CBS).
- **IEEE 802.1Qbv** Time Aware Shaper (TAS).
- **IEEE 802.1Qci** per-stream filtering and policing.
- **IEEE 802.1CB** Frame Replication and Elimination for Reliability (FRER) support.
- **Secure management & security:** HTTPS web UI, SSHv2 CLI, Netconf/YANG, Syslog eventing, encrypted & digitally signed upgrades, 802.1X/RADIUS/RBAC and protocol hardening.



Specifications

SN	Specification	Details
1.	Protocol	Ethernet (IEEE 802.3-2008)
2.	Ports	2x 100/1000BASE-T TSN Ethernet ports; 2x copper SFP modules (10/100/1000) tri-speed RJ45
3.	Host Interface	PCIe interface for attachment to a host computer (1x 1G PCIe port). AXI Stream, AXI lite.
4.	PHY Interface	SGMII
5.	Duplex Support	Full duplex
6.	MAC functions	Automatic MAC address learning and aging; Static MAC table.
7.	VLAN support	Port-based VLANs; IEEE 802.1Q VLAN tagging (up to 4K VLAN groups)

Specifications Cont..

8.	QoS / CoS	IEEE 802.1p Class of Service (CoS) / Quality of Service (QoS)
9.	Discovery / Multicast	IEEE 802.1AB (LLDP); Layer-2 multicast filtering; IGMP snooping (up to 1024 multicast filters)
10.	Traffic control	Port rate limiting; Storm control for flooded broadcast, multicast and unicast
11.	Spanning tree	STP (IEEE 802.1D), RSTP (IEEE 802.1w), MSTP (IEEE 802.1s)
12.	TSN support	IEEE 802.1AS (gPTP), 802.1Qav (CBS), 802.1Qbv (TAS), 802.1Qci, 802.1CB (FRER), 802.1Qcc (SRP enhancements)
13.	Config & Management	HTTPS web interface; SSHv2 CLI; Netconf (YANG) support; encrypted & digitally signed firmware/bitstream upgrades; save/restore configuration; internal monitoring/logging; Syslog event notification; per-port statistics; in-band management via any Ethernet switch port
14.	Security	IEEE 802.1X access control; MAC port binding & authentication; RADIUS authentication; RBAC; selective port disable; disable insecure protocols; HTTPS/SSHv2; encryption/authentication/signature for firmware and bitstream
15.	Card Dimensions	130 mm (W)x180mm(D)x30 mm(H)
16.	Power supply	12V DC (240V AC to 12 V DC power adaptor provided)

Deliverable

1. Qbit TSN End point (SKU:10002)
2. Copper SFPs - 10/100/1000BASE-T SFP SGMII Copper 100m RJ-45 Transceiver Module
3. Ethernet Cables -2
4. Power Adaptor
5. Host PC (windows 11/i7 core)

Get in touch with us

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for Next-Gen Networking

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